

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

April 2013

EE216 Integrated Circuits & Analysis

Date: 27.04.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 What is the meaning of Integrated Circuit (IC)? Give and explain classification of ICs according to their mode of operation. Also explain three basic types of linear IC packages. [07]
- Q - 2 (a) Draw a block diagram and pin diagram of 555 timer IC and also state the function of each pin. [08]
- (b) Draw a diagram of typical 8-pin mini DIP linear op-amp 741 IC including power supply connections. And draw the three different arrangements for positive and negative supply voltages for an op-amp. [06]

OR

- Q - 2 (a) Define the following terms with necessary diagram. [06]
- (i) Common-Mode Rejection Ratio (CMRR)
 - (ii) Slew Rate (SR)
 - (iii) Input bias current
 - (iv) Supply Voltage Rejection Ratio (SVRR)
- (b) Design an astable multivibrator using 555 timer for the output frequency of 5 kHz and 70 % duty cycle. (Take $0.1 \mu\text{F}$) [08]
- Q - 3 (a) Draw and explain second-order low-pass butterworth filter with Frequency response curve. [07]
- (b) Design the first-order high-pass filter with the lower cut-off frequency of 3 kHz. With passband gain of 2. Also draw the frequency response curve. (take $C = 0.01 \mu\text{F}$) [07]

OR

- Q - 3 (a) Draw and explain notch filter with frequency response curve. [06]
- (b) Design the square-wave generator so that $f_o = 1 \text{ kHz}$. The op-amp is a 741 with DC supply voltages $\pm 15 \text{ V}$. (Take $C = 0.1 \mu\text{F}$). Also draw output waveforms. [08]

SECTION – II

Q - 4 Write down the electrical characteristics of an Ideal op-amp. [07]

Q - 5 (a) Draw an equivalent circuit of an op-amp. Also draw and explain its ideal voltage transfer curve. [04]

(b) Design a non-inverting amplifier with closed-loop gain of 3. If input is sine wave of 4 volt peak to peak then draw the output voltage waveforms. [07]

(c) Explain voltage-to-current converter with floating load with neat circuit diagram. [03]

OR

Q - 5 (a) Derive the Slew Rate equation for 741 op-amp with neat diagram. [05]

(b) Using the 7805C voltage regulator, design a current source that will deliver a 0.25 A current to a $48\ \Omega$, 10 W load. [05]

(c) Draw and explain zero-crossing detector with neat waveforms and circuit diagram. [04]

Q - 6 (a) Find out closed-loop voltage gain (A_F) and feedback gain (B) for non-inverting Amplifier with neat circuit diagram. Also draw one-line block diagram for this circuit. [07]

(b) Design a differentiator to differentiate an input signal that varies in frequency from 10 Hz to about 2 kHz. (Take $C_1 = 0.1\ \mu\text{F}$). Draw neat circuit diagram. [07]

OR

Q - 6 (a) The 741 op-amp having following parameter is connected as an inverting amplifier. [07]
with $R_i = 470\ \Omega$ and $R_F = 4.7\ \text{k}\Omega$. Draw circuit diagram and Compute the values of A_F , R_{if} , R_{of} , f_F and V_{OOT}

$A = 200000$, $R_i = 2\ \text{M}\Omega$, $R_o = 75\ \Omega$, $f_o \approx 5\ \text{Hz}$, supply voltages = $\pm 15\ \text{V}$, output voltage swing = $\pm 13\ \text{V}$.

(b) Draw and explain Integrator using 741 op-amp with its frequency response curve. [07]
